

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment & Information Technology

Department of Computer Science

COS326 - Database Systems

Online Examination

Total Marks: 100

14 November 2024

Time/Duration: 7:30am - 10:30am (3 Hours 0 Minutes)

Examiner: Mr. S.M. Makura

External Moderator: Mr F. Elegbeleye (WSU)

Initials and Surname: _____

Student Number: _____

Degree: _____

Instructions

1. Read the questions carefully.
2. This examination comprises of 34 questions on 17 pages. It consists of three sections, namely Section A (Multiple Choice & Fill in the blank questions) – 20 marks, Section B (Application and Theory - Based Questions) – 40 Marks, Section C (Scenario-Based questions) - 40 Marks.
3. You have 180 minutes to complete this examination.

4. This is a **closed book examination**: you may therefore not have any study material with you.
5. Please switch off your smartphone/tablet and keep it off for the duration of the examination.
6. All examination regulations and 'code of conduct' of the University of Pretoria are applicable during this examination.

Integrity Statement:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

Section A: Multiple Choice and Fill in the Blanks Questions (20 Marks)

Question 1 (1 Mark)

You want to create a bi-directional relationship between Student and Course entities, where each student can enroll in multiple courses, and each course can have multiple students. Which annotations and configurations would you use to define the relationship on both sides?

```
@Entity
public class Student {
    @Id
    private Long id;
    private String name;
    // Which annotation would go here?
    private List<Course> courses;
}

@Entity
public class Course {
    @Id
    private Long id;
    private String title;
    // Which annotation would go here?
    private List<Student> students;
}
```

a. @OneToMany on both courses in Student and students in Course.

b. @ManyToMany on courses in Student and @ManyToOne on students in Course.

c. @ManyToMany(mappedBy = "students") on courses in Student and @ManyToMany on students in Course.

d. @ManyToMany on courses in Student and @ManyToMany(mappedBy = "courses") on students in Course.

Question 2 (1 Mark)

A PL/pgSQL function `calculate_discount` applies a discount to a total amount based on customer type. It accepts `customer_id` and `total_amount` as inputs and returns the discounted total. Given the code below, how should you structure the conditional statement to apply a 10% discount for 'VIP' customers and no discount for others?

```
CREATE OR REPLACE FUNCTION calculate_discount(customer_id INT,
total_amount NUMERIC)
RETURNS NUMERIC AS $$
DECLARE
    customer_type VARCHAR;
    discounted_amount NUMERIC;
BEGIN
    -- Fetch customer type
    SELECT type INTO customer_type FROM customers WHERE id =
customer_id;

    -- Apply discount if customer is 'VIP'
    -- Which code from the options correctly applies the discount?
$$ LANGUAGE plpgsql;
```

a. IF customer_type = 'VIP' THEN discounted_amount := total_amount 0.9; ELSE discounted_amount := total_amount; END IF;
b. CASE customer_type WHEN 'VIP' THEN discounted_amount := total_amount 0.9; ELSE discounted_amount := total_amount; END CASE;
c. IF customer_type := 'VIP' THEN discounted_amount := total_amount 0.9; ELSE total_amount;
d. IF customer_type = 'VIP' discounted_amount := total_amount 0.9; ELSE total_amount;

Question 3 **(1 Mark)**

You need to create an XML output that combines the `first_name` and `last_name` of employees in a `Staff` table, with each full name appearing under an XML tag `<FullName>`. Which statement would achieve this?

a. SELECT xmlelement(NAME "FullName", xmlcombine(first_name, last_name))

FROM Staff;

b. SELECT xmlelement(NAME "FullName", xmlconcat(first_name, last_name))

FROM Staff;

c. SELECT xmlelement(NAME "FullName", concat(first_name, last_name))

FROM Staff;

d. SELECT xmlelement(NAME "FullName", xmlattributes(first_name || ' ' || last_name))

FROM Staff;

Question 4 **(1 Mark)**

Given the following inventory collection:

```
{ _id: ObjectId(1), item: "Bike", stock: 20, price: 200 }
{ _id: ObjectId(2), item: "Helmet", stock: 100, price: 50 }
{ _id: ObjectId(3), item: "Gloves", stock: 50, price: 15 }
```

The following aggregation query is run:

```
{ $group: { _id: null, total_stock_value: { $sum: { $multiply: ["$stock", "$price"] } } } }])
```

What is the expected output?

a. { "_id" : null, "total_stock_value" : 5000 }

b. { "_id" : null, "total_stock_value" : 8000 }

c. { "_id" : null, "total_stock_value" : 9750 }

d. { "_id" : null, "total_stock_value" : 4500 }

Question 5 **(1 Mark)**

Suppose you want to update the age of the student named "Andries" to 24 in the `students` collection. Which MongoDB update query will achieve this?

a. db.students.updateOne({ "name": "Andries" }, { $set: { "age": 24 } })

b. db.students.update({ "age": 24 }, { $set: { "name": "Andries" } })

c. db.students.updateMany({ "name": "Andries}, { "age": 24 })

d. db.students.updateOne({ "age": 24 }, { $set: { "name": "Andries" } })

Question 6 **(1 Mark)**

Given the following Cypher query executed in Neo4J:

```
CREATE
```

```
    (s1:Student { name: "Ndivhuwo", grade: "A" }),
    (s2:Student { name: "Marlize", grade: "B" }),
    (s3:Student { name: "Katlego", grade: "A" }),
    (s1)-[:FRIEND_WITH]->(s2),
    (s2)-[:FRIEND_WITH]->(s3),
    (s1)-[:FRIEND_WITH]->(s3);
```

Which Cypher query will return the names of all students who have an "A" grade and are friends with "Ndivhuwo"?

a. MATCH (ndivhuwo:Student { name: " Ndivhuwo " })-[:FRIEND_WITH]->(friend:Student { grade: "A" })

RETURN friend.name;

b. MATCH (ndivhuwo:Student { name: " Ndivhuwo " })-[:FRIEND_WITH]->(friend)

WHERE friend.grade = "A"

RETURN friend.name;

c. MATCH (ndivhuwo:Student { name: " Ndivhuwo " })-[:FRIEND_WITH*]->(friend:Student { grade: "A" })

RETURN friend.name;

d. MATCH (ndivhuwo:Student { name: " Ndivhuwo " })-[:FRIEND_WITH]-(friend:Student { grade: "A" })

RETURN friend.name;

Question 7 **(1 Mark)**

You have a social network graph with nodes representing people and KNOWS relationships between them. You want to find the shortest connection path between "Alice" and "Simphiwe" to see if they are indirectly connected through other friends.

Which query would find the shortest possible path between "Alice" and " Simphiwe" in terms of relationship hops, regardless of relationship direction?

a.

```
MATCH p = shortestPath((alice:Person { name: "Alice" })-[:KNOWS*]-
(simphiwe:Person { name: "Simphiwe" }))

RETURN p
```

b.

```
MATCH p = allShortestPaths((alice:Person { name: "Alice" })-[:KNOWS*]-
(simphiwe:Person { name: " Simphiwe" }))

RETURN p
```

c.

```
MATCH p = shortestPath((alice:Person { name: "Alice" })-[:KNOWS*1..3]-
(simphiwe:Person { name: " Simphiwe" }))

RETURN p
```

d.

```
MATCH p = shortestPath((alice:Person { name: "Alice" })-[:KNOWS*..3]-
(simphiwe:Person { name: "Simphiwe" }))

RETURN p
```

Question 8 **(1 Mark)**

What is the primary advantage of in-memory storage in NewSQL databases?

a. Reduced storage costs

b. Faster query results

c. Enhanced data security

d. Larger storage capacity

Question 9 **(1 Mark)**

What is the main purpose of MapReduce?

a. To compress data.

b. To process large datasets in parallel.

c. To store data in relational databases.

d. To secure data transmissions.

Question 10 **(1 Mark)**

Which of the following is used to create surrogate keys in a ROLAP schema?

a. CREATE SEQUENCE

b. CREATE MATERIALIZED VIEW

c. CREATE DIMENSION

d. CREATE INDEX

Question 11 **(1 Mark)**

The process of combining data and the methods that operate on the data into a single unit in object-oriented databases is called _____.

Question 12 **(1 Mark)**

You have a products table, and you want to create a PL/pgSQL procedure named `delete_product` that deletes a product by its `product_id`. Complete the function below to perform the delete operation.

```
CREATE OR REPLACE PROCEDURE delete_product(product_id INT) AS $$
BEGIN
    DELETE FROM products WHERE _____ = product_id;
END;
$$ LANGUAGE plpgsql;
```

Question 13 **(1 Mark)**

Write out the mongo shell command you would use to delete a collection contained in a particular database in MongoDB [**Note:** write the name of the command only without brackets].

Question 14 **(1 Mark)**

The _____ method in MongoDB returns true if the cursor has documents. [**Note:** write the name of the command only without brackets].

Question 15 **(1 Mark)**

To delete properties and labels from graph elements in Neo4j, you use the _____ clause.

Question 16 **(1 Mark)**

To return unique values from a query and eliminate duplicates, you can use the _____ keyword in Cypher.

```
MATCH (a:Person)-[:FRIEND]->(b:Person)
RETURN _________________ b.name;
```

Question 17 **(1 Mark)**

The ACID properties stand for Atomic, Consistent, Isolated, and _____.

Question 18 **(1 Mark)**

_____ is a Big Data characteristic which refers to the credibility of the source.

Question 19 **(1 Mark)**

_____ refers to a class of applications and technologies used to gather, provide access to, and analyse data about business operations.

Question 20 **(1 Mark)**

In a data warehouse, a _____ change, is when the changed attribute value is overwritten.

Section B: Application and Theory-Based Questions (40 Marks)

Question 21 (3 Marks)

Object ODBMS

Suppose in JPQL you need to query Order entities with a status equal to "shipped". Is "shipped" considered a literal or a variable in the query? Based on your answer write the JPQL query snippet to achieve this. (3 Marks)

Question 22 (4 Marks)

Object Relational DBMS

Differentiate between the third-generation database manifesto and the third manifesto (4 Marks).

Question 23 (5 Marks)

Semi-structured data & XML

Study the following XML document and answer the following question:

```
<catalog>
  <product>
    <name>Book</name>
    <category>Books</category>
  </product>
  <product>
    <name>Laptop</name>
    <category>Electronics</category>
  </product>
</catalog>
```

(a) Write an XPath query to select the name of the product where the category is "Electronics" (2 Marks).

(b) Write an XPath query to select all product elements that have a category of either "Electronics" or "Books" (3 Marks).

Question 24 (7 Marks)

MongoDB

Study the following MongoDB function and answer the following questions:

```
db.orders.aggregate([
    {
        $match: { status: "completed" }
    },
    {
        $group: {
            _id: "$customerId",
            totalAmount: { $sum: "$amount" },
            totalOrders: { $count: {} }
        }
    },
    {
        $sort: { totalAmount: -1 }
    }
])
```

a. Explain in detail what a MongoDB collection is, and what it is composed of (2 Marks).

b. Explain what the MongoDB function above does and describe the role of each stage in the aggregation pipeline (5 Marks).

Question 25 (7 Marks)

Neo4j

Study the following CYPHER query and answer the questions that follow:

```
CREATE (jane:Customer {name: 'Jane', age: 32})
CREATE (virat:Customer {name: 'John', age: 45})
CREATE (tv:Product {name: 'TV', price: 300})
CREATE (phone:Product {name: 'Smartphone', price: 700})

CREATE (jane)-[:PURCHASED]->(tv)
CREATE (virat)-[:PURCHASED]->(phone)
```

a. What does the [:PURCHASED] relationship represent in this scenario? Write the Cypher query to retrieve all customers who purchased a TV? (5 Marks)

b. Write a Cypher query to find the total number of products purchased by customers. (2 Marks)

Question 26 **(7 Marks)**

Big Data and Data Mining

(a) Compare and contrast parallel relational DBMSs and the MapReduce/Hadoop systems (4 Marks)

(b) Elaborate on at **least two** ways to describe the knowledge discovered during data mining (no one word answers). (3 Marks)

Question 27 **(7 Marks)**

Data Warehousing and OLAP

(a) What is a Fact Table in a Data Warehouse? Describe its role in OLAP operations. (3 Marks)

(b)Perform a Drill-Down OLAP operation on the following quarterly sales data to show monthly sales figures. Assume the total sales for each quarter is equally distributed among the months in that quarter. (4 Marks)

Year	Quarter	Total Sales (R)
2023	Q1	15,000
2023	Q2	20,000
2023	Q3	25,000
2023	Q4	30,000

Section C: Scenario-Based Questions (40 Marks)

Question 28 **(4 Marks)**

Object ODBMS

Study the following scenario and the following question:

You are building a system for an online marketplace where each **Product** can be sold by multiple **Sellers**, and each **Seller** can sell multiple **Products**. You need to

ensure that the relationship is **bidirectional** so that you can easily navigate from either **Product** to **Seller** or from **Seller** to **Product**.

Source: Makura S.M (2024)

(a) How would you implement this bi-directional relationship using JPA annotations, ensuring that both entities maintain a reference to each other without causing issues like infinite loops during serialization? (3 Marks)

Question 29 (4 Marks)

Object RDBMS

Study the following SQL statements and answer the question which follows.

```
CREATE TYPE VehicleType AS ENUM ('Car', 'Truck', 'Motorcycle');
CREATE TYPE OwnerType (ownerId INT, ownerName CHAR(50), vehicle
VehicleType);
CREATE TABLE vehicleOwner (owner OwnerType, registrationDate DATE);
```

Source: Makura S.M (2024)

Write a PL/pgSQL function called OwnerDetails that returns a string with the owner's ID, name, and the type of vehicle they own (4 Marks).

Question 30 (5 Marks)

Semi structured Data & XML

You are developing a sales database for a retail company that tracks customer purchases and their corresponding product details. The database stores customer information, purchase history in XML format, and an array of product IDs for each order.

The following SQL statements were successfully executed in PostgreSQL:

```
CREATE TABLE CustomerSales (
    customerId CHAR(5) PRIMARY KEY,
    customerName VARCHAR(100),
    purchaseHistory XML,
    productIds INT[]
);
```

```
INSERT INTO CustomerSales (customerId, customerName, purchaseHistory,
productIds)
VALUES ('C001', 'Alice Johnson',
    XML('<PURCHASES>
            <PURCHASE date="2024-09-10">
                <PRODUCT>101</PRODUCT>
                <AMOUNT>150.00</AMOUNT>
            </PURCHASE>
            <PURCHASE date="2024-09-12">
                <PRODUCT>102</PRODUCT>
                <AMOUNT>200.00</AMOUNT>
            </PURCHASE>
        </PURCHASES>'),
    ARRAY[101, 102]);

INSERT INTO CustomerSales (customerId, customerName, purchaseHistory,
productIds)
VALUES ('C002', 'Bob Smith',
    XML('<PURCHASES>
            <PURCHASE date="2024-09-11">
                <PRODUCT>103</PRODUCT>
                <AMOUNT>120.00</AMOUNT>
            </PURCHASE>
        </PURCHASES>'),
    ARRAY[103]);
```

Source: Makura S.M (2024)

Write a query to extract the total purchase amount from the purchaseHistory XML field for each customer using the xpath() function. [5 Marks]

Question 31 (7 Marks)

MongoDB

Study the following scenario and the following questions:

You are a database manager for WildlifeSA, an organization dedicated to tracking and protecting endangered wildlife species across South Africa. WildlifeSA maintains a MongoDB database called **conservationDB**, which includes a collection named **animalSightings**. This collection stores records of animal sightings, including species, location, date, ranger details, and any special notes. Each document in the **animalSightings** collection has a structure like this:

```
{
  "_id": 1,
  "species": "African Elephant",
  "location": "Kruger National Park",
  "date": "2024-01-15",
```

```
  "ranger": { "name": "Sipho Dlamini", "rangerID": "RD123" },
  "notes": ["Near waterhole", "Healthy", "In herd"]
}
```

Source: Makura S.M (2024)

Write MongoDB queries to retrieve specific details from the animalSightings collection:

a. Find and display only the species and location of sightings recorded in January 2024. (3 Marks)

b. Retrieve the complete details of any sighting recorded by ranger "Nomsa Khumalo". (2 Marks)

c. After analysing data, the conservation team decides to delete all records of sightings where the species is not endangered. Write a JavaScript statement to remove any documents from the **animalSightings** collection where the species is "Impala". (2 Marks)

Question 32 (7 Marks)

Neo4j

Study the following scenario and the following questions:

You are working for HeritageSA, an organization that tracks South African heritage sites and famous personalities associated with them. The Neo4j database contains a collection of nodes and relationships that capture these connections. The following Cypher query was used to set up part of the dataset in the database:

```
CREATE
      ( mNelson:Person { name: "Nelson Mandela", born: 1918, role:
"Former President" }),
      ( mTutu:Person { name: "Desmond Tutu", born: 1931, role:
"Archbishop" }),
      ( cCapeTown:City { name: "Cape Town", province: "Western Cape" }),
      ( sRobbenIsland:Site { name: "Robben Island", type: "Heritage Site"
}),
      ( sApartheidMuseum:Site { name: "Apartheid Museum", type: "Heritage
Site" }),

      ( mNelson)-[:VISITED { year: 1994 }]->(sRobbenIsland),
      ( mNelson)-[:HONOURED_IN { year: 2001 }]->(sApartheidMuseum),
      ( mTutu)-[:HONOURED_IN { year: 2010 }]->(sApartheidMuseum),
      ( cCapeTown)-[:HAS_SITE]->(sRobbenIsland);
```

Assuming that it has been already executed in a Neo4j database, use this query to answer the following questions.

Source: Makura S.M (2024)

(a) Complete the following Cypher query to update each Site node with a new property called **visitorsPerYear** and assign values of 300,000 for Robben Island and 150,000 for the Apartheid Museum (2 Marks).

```
MATCH (sRobbenIsland:Site)
WHERE sRobbenIsland.name = "Robben Island"
SET ____________________
RETURN sRobbenIsland;

MATCH (sApartheidMuseum:Site)
WHERE sApartheidMuseum.name = "Apartheid Museum"
SET ____________________
RETURN sApartheidMuseum;
```

(b) Write a Cypher query to find all the people honored in the Apartheid Museum and return their names and the year they were honored (2 Marks).

(c) Write a Cypher query to retrieve the names of all heritage sites located in Cape Town (3 Marks).

Question 33 (7 Marks)

Data Mining

Study the following scenario and the following question:

You are working as a Data Mining Specialist at Smart Shop supermarket and have been given the following dataset:

Transaction ID	Items Purchased
101	Milk, Butter, Bread
102	Milk, Bread
103	Butter, Bread, Sugar
104	Milk, Butter
105	Milk, Bread, Sugar
106	Butter, Sugar

Set of Items: {Milk, Butter, Bread, Sugar}

Source: Makura S.M(2024)

Apply the Apriori algorithm to the above dataset. Use 33% for the minimum support value. Show all your working. (7 Marks)

Question 34 **(7 Marks)**

Data warehousing and OLAP

Study the following scenario and the following questions:

You are working as a business analyst at MiCasa Co. MiCasa Co is a local company in South Africa with hotels in four cities in South Africa. After receiving the profits results for the year 2022 from each of the branch managers of the store, you perform the following operations in the Datawarehouse:

```
CREATE TABLE HOTEL ……;
CREATE DIMENSION HOTELS_DIM
    LEVEL Hotel  IS HOTEL.Hotel_ID
    LEVEL Province IS HOTEL.Province_ID
    LEVEL Hotelline IS HOTEL.Hotelline_ID

HIERARCHY Hotel_Rollup
    (Hotel      CHILD OF
    Province CHILD OF
    Hotelline)

ATTRIBUTE Hotel DETERMINES
          (Hotel_name, Hotel_address, Hotel_city……)
ATTRIBUTE  Province DETERMINES Province_name
ATTRIBUTE Hotelline DETERMINES Hotelline_desc
```

Source: Makura S.M (2024)

a. Write the SQL statement to create the TIME dimension. This dimension should have a hierarchy that rolls up as follows: DAY -> MONTH -> QUARTER -> YEAR (4 Marks)

b. Write the SQL statement that will compute the total revenue for each hotel, for each month. Use a GROUP BY CUBE. Explain all the different values you expect to see in the output. (3 Marks)

End of examination paper